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A survey on FOG computing for Smart waste management system

Pooja V. Garach
Computer Engineering Department
Silver Oak College of Engineering
Ahmedabad, India
pvgarach@gmail.com

Rikin Thakkar, Assistant Professor
Computer Engineering Department
Silver Oak College of Engineering
Ahmedabad, India
rikinthakkar1991@gmail.com

***Abstract*-Since the invention of 'Cloud computing' in the year 2009, with the help of its three essential attributes i.e. Software as a Service (SaaS), Platform as a Service (PaaS) and Infrastructure as a Service (IaaS), the technology is being has been widely used for the conventional data storage, running applications and works as an operating system as well. However, due to more latency time, undefined security and centralized geo-distribution, support of mobility is limited in cloud computing. With the need to overcome these drawbacks, a new platform has been evolved over a period of time named as 'Fog Computing', which already in use for last few years in advertising, computing, entertainment, data analysis and distributed data collection purpose. Extending the usability of its unique features, it has been assessed that for the sub-systems like 'Smart waste management' of 'Smart City' concept, 'Fog' can be implemented with the greater ease of operation, efficient management of system and great improved latency time. With the use of optimised algorithm, an effective, reliable, cost effective and feasible system can be implemented as an effort towards realizing the smart waste management system, specifically for the 'Smart City' concept. The present survey paper is to assess the various aspect of utilizing the 'Fog' as primary tool as an advancement to the conventional 'Cloud' based waste management system and check the feasibility of its implementation taking into account the corresponding experiences in this area.**

Keywords—Cloud computing;Fog computing, Smart waste management system

I. INTRODUCTION

Sustainable development of human being on the earth depends on 'how a safe and clean environment' is available and thereby maintaining the hygiene in the day to day life. As a dark side of industrial revolutions across the globe, the ecological system on earth is hampered to a large extent. This has resulted in increased risk to not only human life but also to the other living creatures. Keeping the reasons aside for these disasters, which is altogether a different subject, it is now high time that we concentrate on the managing our ecosystem properly and invest a part of our time, effort and money into it. One of such areas, where the attention is required, is the handling of tons of waste produced by some or other means. Unless they are properly collected and handled upto the point of biodegradable level, they are damaging the surrounding environment and related living elements. With the advancement of technologies in many areas, a concept called 'smart' is evolved over a period of last few decades across the world and became popular in India now in 21st century. As a part of 'smart city' concept, 'smart waste management' is emerged as one of the major area which needs to be developed and implemented utilising various technologies those are available.

Toward the contribution in the initiatives undertaken by government in this area, we are aiming to design and develop such a smart waste management system which is efficient, reliable, time saving, cost effective and feasible to apply for the lowest link of the society. While doing so, it is important to take a stock of what is being practiced around the world, especially in the developed countries, thoroughly study them and then integrate the best possible elements of their system to the latest technology options.

Essential element of this system is the data server which can handle huge data followed by its processing and provide feedback accordingly. 'Cloud' is such a data server, which is widely used for this purpose. If we go by the strict definition of 'cloud' as defined in literature, cloud is a large group of interconnected computers and these computers can be personal computers, network servers[26]. Cloud can be single or multiple and broadly classified as Private, Public and Hybrid Cloud.

The next chapters of this paper describe the review and survey of such systems in form of background studies (chapter II) which have gone into the development of these system and what we can do for our application. Further, the key features, improvement possibilities and drawbacks of existing systems have been described in chapter III.

II. BACKGROUND THEORY

A. Fog Computing

Fog computing is a distributed computing model acting as an intermediate layer between Cloud datacentres and IoT devices/sensors offering storage facilities ,compute, networking so that Cloud-based services can be extended closer to the IoT devices/sensors[7]. The concept of Fog computing was first introduced by Cisco in 2012 to address the challenges of IoT applications in conventional Cloud computing[6].

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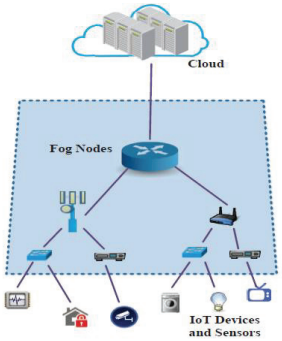

Fig. 1. Fog Computing Environment [6]

Environment of Fog computing is comprised of traditional networking components such as switches, proxy servers ,routers, set top boxes, Base Stations, etc. and can be located nearer to IoT devices/sensors as shown in Fig.1 which are provided with different storage, networking ,computing capabilities. The components of Fog computing enables to create huge Cloud-based services geographically, additionally it provides facility for scalability ,location awareness, mobility support , interoperability, real-time interactions[5]. Fog computing performs efficiently in network traffic, capital and operational expenses, service latency, content distribution ,power consumption etc. and hence it proves better than simply usage of cloud services only[6] .

Extension of cloud is named as 'Fog', which produces and act on IoT data. These devices are known as fog nodes that can be used anywhere in network connection such as alongside a railway track, on a factory floor, in a vehicle, on top of a power pole or on an oil rig. Thus, fog node is a device that can compute, store and has a network connectivity such as video surveillance cameras, industrial controllers, switches, embedded servers, routers[3]. Cloud and Fog are connected using Fog nodes with cloud platform.

Following functions are performed by the Fog nodes:

- Fog nodes receives data from IoT device in real time using protocols
- Running IoT application with lesser response time
- Providing 1–2 hours of transient storage
- Send summary of data to the cloud

Whereas the cloud platform:

- Receiving data summary from the fog nodes
- Performing analysis on data made available from IoT devices gaining business insight
- Sending new protocols to the fog nodes depending on these insight

Major advantages of Fog computing includes (i) reduction of network traffic (ii) suitable for IoT tasks and queries (iii) low latency requirement (iv) scalability [7].

B. Fog Architecture

Figure 2 shows a Fog computing architecture. Layer 4 consist of the sensing network having sensory nodes which are invasive, highly reliable, and low cost, generating exceptionally larger sensing data streams from sensors which have to be processed. Layer 3 receives the raw data from the node from previous layer. It is made up of many high-performance and low-power computing nodes or edge devices which forms a small group of sensor executing the analysis of data periodically. The output of the nodes is the reports of data processing results and quick feedback control to a their devices for its monitoring. In layer 2 there are lots of intermediate computing nodes connected to the edge devices of layer3 associating spatial and temporal data for harmful events. Also, the immediate feedback control is provided for layer 2, layer 3 and all the analysis of data is submitted to top most layer for better analysis and monitoring. The uppermost layer is the Cloud Computing data center that provides centralized monitoring and controlling. It has large capacity for complex task performing which support dynamic decision making. Thus, Fog Computing architecture has high performance computing power with intelligence for developing smart city[27].

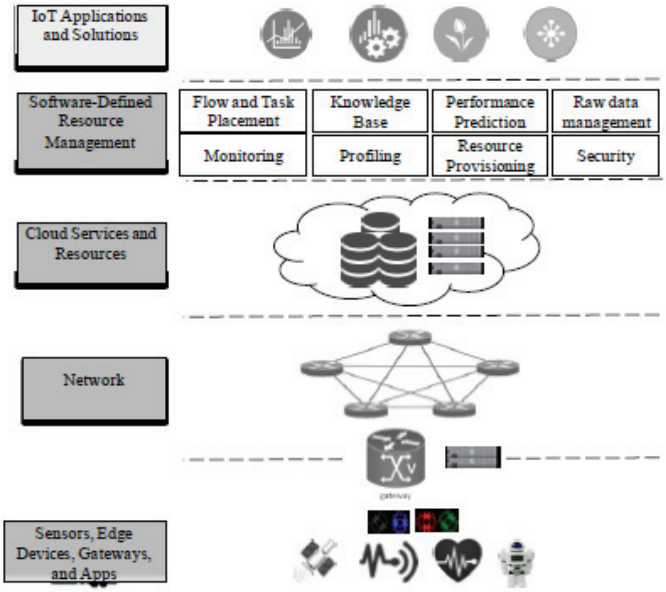

Fig. 2. Fog computing architecture [27]

The Fog Computing is benefited as greater business agility with the right tools in which machine manufacturers offers machine-as-a-service to their customers, better security protecting the fog nodes, deeper insights, with privacy control where privacy and secrecy are maintained, Lower operating expense where by conserving bandwidth of network [3].

C. Applications of Fog computing

Numerous applications benefiting from Fog computing paradigm are as follows [3][4] [7]:

- Smart Healthcare system[7]
- Augmented Reality where EEG sensors are generated and classified to detect the brain state of the player by collecting raw stream of data[7]

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- Caching and Pre-processing is carried using edge servers for improving web sites performance[7]
- A Smart Traffic Light System[4]
- A Wind farm[4]
- Locking a door [3]
- Smart home[3]
- Smart grid[3]
- Smart vehicle[3]
- Changing equipment settings[3]
- Applying the brakes on a train[3]
- Zooming a video camera[3]
- Opening a valve in response to a pressure reading[3]
- Sending an alert to a technician to make a preventive repair[3]

D. SMART WASTE MANAGEMENT

Smart waste management is the system defined for the Smart city concept. In the smart waste management system as the level of garbage in the bins cross the threshold, it will be informed to the corresponding authority, using the IOT device. At the first level of this process, clustering is done using k-means clustering technique in which nearby bins belonging to similar region cluster is formed and according to that shortest path is define. Authority will then inform truck driver to collect the waste from the filled bins according to the shortest path formed from the data retrieved and those waste are disposed at particular location.

Smart waste management system basically deals with the garbage monitoring, collection and disposing of the garbage. Various hardware tools are used for the waste monitoring such as RFID, Arduino UNO, Raspbery Pi, Wireless Sensor network[10],[12],[13],[15],[16]. For the collection of waste in the wastes bin route has been define in such a way that it form the shortest path from source to destination which will be cost effective and time efficient .

The following are the algorithm which are used to find the shortest path which are used in one or the other system.

- Decision Support System (DSS) [12]
- Dijkstra's algorithm for computing shortest path[15][18]
- Floyd-Warshall algorithm[15]
- Traveling Salesman Problem(TSP)[12][13]
- Top-k query processing [12][15]
- Genetic Algorithms (GA)[16][34]
- K-means Clustering [16]
- Nearest-neighbour shortest path algorithm for finding the optimized routes[17]
- Shortest path spanning tree algorithm (SPST)[17]
- Ant Colony Optimization(ACO)[19]

III. A SURVEY ON ALGORITHM IMPLEMENTED FOR SMART WASTE MANAGEMENT

A .Dijkstra's Algorithm

Dijkstra's algorithm finds the length of an shortest path between two vertices in a graph. The algorithm works as follows:

1. Select source vertex
2. Define a set S of vertices, and initialise it to empty set. As the algorithm progresses, the set S will store those vertices to which a shortest path has been found.
3. Label the source vertex with 0, and insert it into S.
4. Consider each vertex not in S connected by an edge from the newly inserted vertex. Label the vertex not in S with the label of the newly inserted vertex + the length of the edge.
But if the vertex not in S was already labelled, its new label will be min(label of newly inserted vertex + length of edge, old label)
5. Pick a vertex not in S with the smallest label, and add it to S.
6. Repeat from step 4, until the destination vertex is in S or there are no labelled vertices not in S.

If the destination is labelled, its label is the distance from source to destination. If it is not labelled, there is no path from the source to the destination [23].

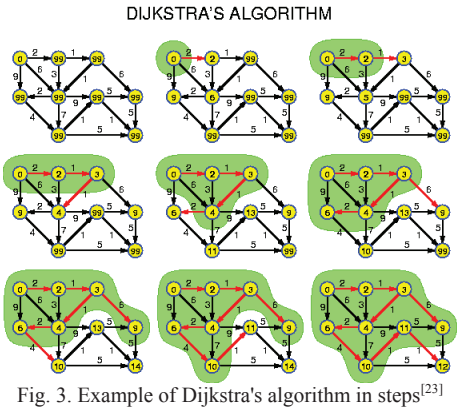

Fig. 3. Example of Dijkstra's algorithm in steps[23]

B. Ant Colony System

With an inspiration from the nature, a unique algorithm have been developed during the last decade of 20th century, named as Ant Colony Algorithm (ACA), which adopts the brilliant methodology of real ants for exchanging the information via pheromone deposited on the ground. This system is already implemented and running successfully for Travelling Salesman Problem (TSP), Quadratic Assessment problem and job-shop scheduling. Looking at the versatility, robustness of the theory and practically proven technique, it became an obvious choice for the researchers to assess its implementation possibility for the recent problem like smart waste management[35].

A typical Ant colony algorithm can be described as follows:

1. Initialize:
   Set t :== 0 {t is the time counter}

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Set NC := 0{NC is the cycles counter}
For every edge (i, j) set an initial value Tij (t) = c for trail intensity and I:!.Tij = 0
Place the m ants on the n nodes

2. Sets:= 1 {s is the ants list index}
For k := 1to m do
Place the starting town of the kth ant in $ants_k$ (s)

3. Repeat until ants list is full
{this step will be repeated (n - 1) times}
Sets:= s + 1
For k := 1to m do
Choose the town j to move to, with probability pt(t) given by Eq. (4)
{at time t the kth ant is on town i= antsk(s- 1)}
Move the kth ant to the town j
Insert town j in $ants_k$ (s)

4. For k := 1 to m do
Move the kth ant from $ants_k(n)$ to $ants_k$ (l)
Compute the length Lk of the tour described by the kth ant
Update the shortest tour found For every edge (i, j)
For k := 1 to m do

$$\Delta\tau_{ij}^k = \begin{cases} \frac{Q}{L_k} & \text{If (i,j) E tour described by ants}_k \\ 0 & \text{otherwise} \end{cases}$$

$$\Delta\tau_{ij} := \Delta\tau_{ij} + \Delta\tau_{ij}^k;$$

5. For every edge (i,j) compute TiJ(t + n) according to equation Tij(t + n) = p ·Tij(t) + !:c.Tij
Set t := t + n
Set NC := NC + 1
For every edge (i,j) set !:c.Tij :=0

6. If (NC < NCMAx) and (not stagnation behaviour) then
Empty all ants lists
Goto step 2
Else
Print shortest tour
Stop

Where ants list is define as the list where an ant that visit different towns and associating it with data structure. The difference of the elements recorded in the ants list, various permutations in search cases, nodes in the AS. Here antsk is the kth ant, the set obtained from the elements of antsk, and antsk(s) the sth element of the list

C. Floyd–Warshall Algorithm

One of the very old concept to find the shortest path in a weighted path (positive or negative edge weights) was to compare all the possible path through the graph between each pair of vertices, popularly known as Floyd-Warshall algorithm. It is a classic example of dynamic programing, where this is achieved by $\Theta(|V|^3)$ comparison in a graph. Considering the fact that there could be up to $\Omega\ |V|^2$ [36].

```
for each vertex v
   d[v][v] ← 0
for each edge (e, v)
   d[u][v] ← w(u,v)
for k ← 1 to n do
   for i ← 1 to n do
      for j ← 1 to n do
         if d[i][j] > d[i][k] + d[k][j]
            d[i][j] ← d[i][k] + d[k][j]
         end if
```

IV. LITERATURE SURVEY

In paper [10], for the intelligent garbage collection, a smart alert system has been propose wherein a alert signal, generated through an Ultrasonic sensor (interfaced with Arduino UNO), is provided to municipal web server. On receipt of such alert, the driver visits the particular location and performs the task of emptying the dustbin. In this process RFID is being used for auto-detection of status of dustbin i.e. empty or filled. Once the task of emptying is done, signals are sent back to server about the accomplishment of the task. An integrated module with RFID and IOT has been designed and proposed in this work.

In paper [11], the mechanism to prioritise the collection based on the location e.g. schools or hospitals has been integrated together and in that way a dynamic waste management system has been proposed. Further, the similar priorities has been identified for the dangerous waste (i.e. causing the quick health impact to people living areas). The mentioned goal is achieved by means of novel algorithms which optimises the priority and related cost. In current method data is evaluated with real time and synthetic data is retrieved by municipality of Saint Petesburg, Russia and for this they have designed and developed models like dedicated trucks model, detour model, minimum distance model, and reassignment model.

In paper [13] , the goal defined here is to reduce the power consumption and increase the operational time by designing a system which collects the data and deliver the data through wireless mesh network. The architecture considered for this expands over three tier namely, outdoor nodes (to sense the fullness of bin through sensor node), analytics (to analyse the data, process them, tag the metadata and then interface with the external system) and workstation (which works as the graphical interface for user). This system operates with the data delivery ratio of 99.25% and can be effectively used for litter bin daily seasonality information.

In paper [14], the present system is configured as 'pre-separated waste' for differentiating the database of waste collected obtained from the sensor with respect to its category i.e. organic, plastic, paper, bottle, metal etc. This enables to have an efficient waste management system and has been adopted in Korea. In the description of such a system, a generic work flow has been provided, wherein, on receipt of alert message, collection is to be arranged and once the task is done the status in the system is updated accordingly. While the types of smart bins and cloud architecture is configured, the

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real implementation methodology which takes care of the different variables of the system has not been described.

In paper [15] ,with the additional incorporation of 'real-time' information on the status of the bin, expected fill-up time of the bins, and level of harmful gases, the IoT system proposed here can be employed by municipal administration for society level waste collection system. In the architecture, it comprise of devices embedded with microcontroller , sensors and communicator to the workstation (located at municipal office). However, the algorithm based shortest path determination and thereby saving on cost and time has not been exercised.

In paper [16],with the primary focus on economical and efficient waste collection system, this paper has described the comparative study with sectorial waste collection system. The system implementation is through Shortest Path Spanning Tree (SPST), Generic Algorithm (GA) and k-means algorithms. They have conducted experiments on geographical information, system simulation environment and graph optimization algorithm. The observation shows that, it takes only ~2 years of time to recover the expenses incurred in implementation of such intelligent solution.

In paper [17] , a simulation study has been carried out for a wider area (~10 cities and 5000 bins) with consideration of the reduction in length of route, time to cover that route and cost of collection and disposal of wastes, as the performance parameters. The system has an advantage of handling huge volume of data through internet. The same further can be put into spatio-temporal context and processed by intelligent and optimised algorithms like Smart Waste Management Algorithm, Shortest path spanning tree algorithm, Algorithm in wastebin sensors, Analysis algorithm used by the sensors. Though the actual numbers on how effectively the performance parameters have been achieved to the level of expectation have not been calculated, it provides a good methodology to integrate various algorithm to put into application of such kind.

In paper [19],with the central monitoring of the waste management system, the presented method optimises the shortest possible collection path using Ant Colony Optimization (ACO) technique. The flexibility in the system is also incorporated which allows to change the route in case of blockage of roads. The process works in two phases: In first phase the data is collected from waste bins through economical sensors and the same are being sent to cloud server using micro-controller. In second phase, the optimal route is derived by above mentioned technique. In this phase, the mining usage pattern and weight sensor data at different time is also helpful to define the optimal route.

In paper [20] ,unlike the central monitoring system, the information here directly reaches to cleaner's mobile using Wi-Fi module connected to Raspberry pi. The bins are equipped with ultrasonic sensors with interface of Arduino UNO to check the level of garbage bins. The data collection is in the cloud with the help of Ubidots Software. To aid the system, real time monitoring and follow-up can be adopted by municipal authorities. The proposed system is aimed to have cost reduction, resource optimization, effective use of smart dustbins and seamless operation.

In paper [29] it shows architecture, modelling, simulation, and implementation of dynamic system for common user which transmit fill level of waste bins consuming less power and having low-cost embedded components. The sensing units comprised of ultrasonic sensors send fill level based on MATLAB simulation. Main part of this paper is RFID system which controls the sensors and sends the reading data from the receiver.

In paper [30], presents work based on "Swachh Bharat Abhiyaan" with usage of Smartphone &sensors in embedded system to categorize wet and dry waste. This paper describes technique for collection of dry and wet waste placed in conveyor belt which sends signal to Microcontroller having RF technology . Using IoT module controls and monitor waste information which is send to controlling organization and who collects the waste accordingly.

In paper [31], they have proposed smart waste collection system which requires interconnection of dissimilar devices and data sharing so this can be solved by Smart-M3 platform offering a higher scalability and decoupling which exempt from collecting semi-empty bins. Also this method provides decisional algorithm to obtain most available number of waste vehicles or bins that can locate in nearby area. Their methods proves advantageous to providers and consumers which is cost effective.

V. EXPERIMENTAL SETUP

Our current system works on the waste management where it collects data from the waste bins from different areas. Here word smart waste management is used as the smart means automated system in which a ultrasonic sensor located inside the lid or on the the internal and upper part of the bin; and a weight sensor in the bottom of the container. The ultrasonic sensor measures the level of fullness of the container; the weight sensor measures the amount of trash contained in the bin or thrown by a user. Each and every bin is connected with Arduino node MCU module which will send the data to the to the fog node using wifi inbuilt in it . This in turn sends alert message to the garbage collector for collecting garbage as an when the bins get filled up. The garbage collector of that area will receive shortest path route for the collection from the Fog nodes.

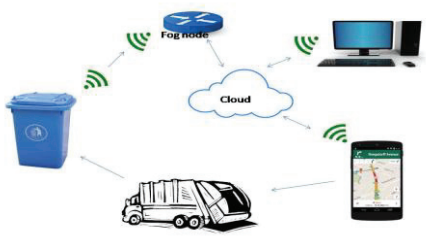

Fig. 1. Experimental setup for smart waste management

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VI. OPEN ENDED PROBLEMS

Research can be done in this field in various aspects such as smart traffic light system, wind farms[3] . Fog computing grows drastically with IoT, radio access techniques, network function virtualization, , traffic monitoring, programmable interfaces, software-defined networking and Mobile cloud and Virtual Machine migration [28].

VII. CONCLUSION

With the mandate of Swatch Bharat Abhiyan' defined by Govt of India, various smart waste management systems have been thoroughly reviewed as discussed. Each of them have been evaluated with respect to their merits and possible improvements along with the feasibility of practical implementation. Based on the above, an approach towards dynamic, feasible, faster, responsive and practical solution in support of the government effort has been proposed here.In this paper we described a survey for smart waste management system methods and effective ways for hygienic environment from various location where they have try to implement this system or it is in execution. For smart waste management either they have used Arduino Uno or wireless sensor along with Raspbery Pi or wifi module and storing and retrieving data from cloud or from fog devices. This survey is helpful for making this system for its effective implementation for making city clean and green.

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